Employee Attrition Prediction Using Machine Learning With Python

is a Course Based Project in partial fulfillment of the requirements for the

Course name: Machine Learning with Python
Course code: E1PY210B
Program: Python
Semester: II
Section: 14

SUBMITTED BY:

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SUBMITTED TO:

SCHOOL OF COMPUTER APPLICATIONS & TECHNOLOGY (SCAT)

DECLARATION

I hereby declare that the project report entitled "Employee Attrition Prediction" submitted to **School of Computer Applications & Technology, Galgotias University** in partial fulfillment of the requirement for the award for the degree of **Master of Science (M.Sc.) in Data Analytics,** is an authentic and original work carried out by me

The matter embodied in this project is a genuine work done by me and has not been submitted whether to this institute or to any other University / Institute for the fulfillment of the requirements of any course of study.

Wherever I/We have used materials (data, mathematical analysis and text) from other sources or have quoted written materials, I have given due credit to them by giving their details in the references section.

Name	Enrollment No.	Role	Sign.
Abdillahi Mahdi Ibrahim	25scse2160065	Project Leader & Machine Learning Developer	

(Signature of approving Faculty Member)

Ms./Mr./Dr. …….…………………….
(Emp ID)…………………….………...
(Affiliation)………………….………...

ACKNOWLEDGEMENT

First of all, I would like to pay my humble respect to the Almighty God for His grace and mercy, by which I was able to complete this project.

I would like to express my heartfelt gratitude to Prof. (Dr.) Vineeta Khemchandan, Dean, School of Computer Applications & Technology, Galgotias University, for the profound and continuous support provided throughout this project.

Further, I express a deep sense of gratitude to Dr. Jha Gautam Kumar

Program Chair – M.Sc. Data Analytics, School of Computer Applications & Technology, Galgotias University, for their cordial guidance and support and for making available all the required resources and materials necessary to complete this project.

I would like to extend my sincere gratitude to Ms./Mr./Dr. Jha Gautam Kumar, [Affiliation/Designation], School of Computer Applications & Technology, Galgotias University, for valuable guidance, constructive suggestions, and continuous support in completing this project.

I would also like to express my sincere thanks to Mr./Dr. Siddharth Dhirendra,

Lab Instructor, School of Computer Applications & Technology, Galgotias University, for their technical assistance, valuable suggestions, and support during the practical implementation of this project.

I also acknowledge the encouragement and support of my parents, fellow students, friends, family members, and all those who were directly or indirectly associated with the successful completion of this project.

Date: ………………….	Name:- …...………………………
Signature ..………………………	Enrollment No. …………….

Table of Contents

Contents

Table of Figures

Chapter- 1

Introduction (Preliminary Project Plan)

Employee attrition refers to the phenomenon whereby employees voluntarily or involuntarily exit an organization through resignation, retirement, or termination. High attrition rates present a significant operational and financial challenge, leading to increased recruitment costs, diminished productivity, and loss of institutional knowledge.

This project leverages machine learning algorithms to predict whether an individual employee is at risk of leaving the organization. By proactively identifying high-risk employees, HR departments can implement targeted retention strategies, thereby reducing overall turnover and its associated costs.

The dataset used in this project is sourced from Kaggle and contains a comprehensive set of employee attributes including age, monthly income, department, job role, years of experience, and satisfaction levels. These features collectively serve as predictors for the target variable — employee attrition.

i. Problem statement

Employee attrition is a persistent challenge across industries. The unplanned departure of skilled employees disrupts workflows, burdens remaining staff, and escalates costs associated with hiring and onboarding replacements.

This project addresses the need for a data-driven solution: developing a machine learning model capable of accurately predicting whether an employee will leave the organization based on historical employee data. The resulting system is intended to support HR professionals in making informed, evidence-based workforce decisions.

ii. Objective

The key objectives of this project are:

- Conduct thorough analysis of employee-related data to uncover trends and patterns.
- Identify and quantify the key factors contributing to employee attrition.
- Preprocess and clean the dataset to ensure model reliability.
- Build and train a robust machine learning classification model.
- Evaluate model performance using standard metrics including accuracy, precision, and recall.
- Provide actionable insights to help organizations design effective employee retention programmers.

iii. Proposed Methodology

The project follows a structured, six-stage machine learning pipeline:

01	**Data Collection** Download the Kaggle Employee Attrition Dataset and load it into the Python environment using Pandas.
02	**Data Preprocessing** Handle missing values, remove duplicate records, encode categorical variables using Label Encoding, and scale numerical features.
03	**Exploratory Data Analysis (EDA)** Analyses employee demographics, visualise attrition distributions, and identify the most influential features.
04	**Model Building** Split the dataset into training (80%) and testing (20%) subsets and train classification models.
05	**Model Evaluation** Compute accuracy, generate a confusion matrix, and compare model performance across algorithms.
06	**Prediction** Apply the trained model to new employee records to generate attrition predictions.

iv. Expected Outcome

Upon successful completion of this project, the following outcomes are anticipated:

- Accurate classification of employees as likely or unlikely to leave.
- Clear identification of the features that most strongly influence attrition.
- Enhanced workforce planning capabilities for HR departments.
- Development of data-driven employee retention strategies.
- A reusable, extensible codebase suitable for integration into broader HR systems.

v. Future-scope and Limitations

5.1 Future Scope

- Deploy the trained model as an interactive web application using Flask or Streamlit.
- Integrate with live HR information systems to enable real-time attrition monitoring.
- Extend the model with advanced algorithms such as XGBoost, LightGBM, or deep neural networks.
- Develop dedicated HR dashboards providing visual, decision-ready insights.
- Implement automated alert systems to notify managers when employee risk scores exceed defined thresholds.

5.2 Current Limitations

- Model accuracy is contingent on the completeness and quality of the underlying dataset.
- Findings may not generalise uniformly across all industries or organisational contexts.
- A constrained feature set may limit the model's predictive depth.
- Historical data may not fully capture emerging workforce dynamics or cultural shifts.

Chapter- 2

Technical Requirements & Design

1. Libraries and Dependencies

The following Python libraries are required for this project:

Library	Version	Purpose
Pandas	Latest	Data manipulation and analysis
NumPy	Latest	Numerical computing
Matplotlib	Latest	Static data visualisation
Seaborn	Latest	Statistical data visualisation
Scikit-Learn	Latest	Machine learning algorithms and utilities
Joblib	Latest	Model serialisation and persistence

Installation Command

```
pip install pandas numpy matplotlib seaborn scikit-learn joblib
```

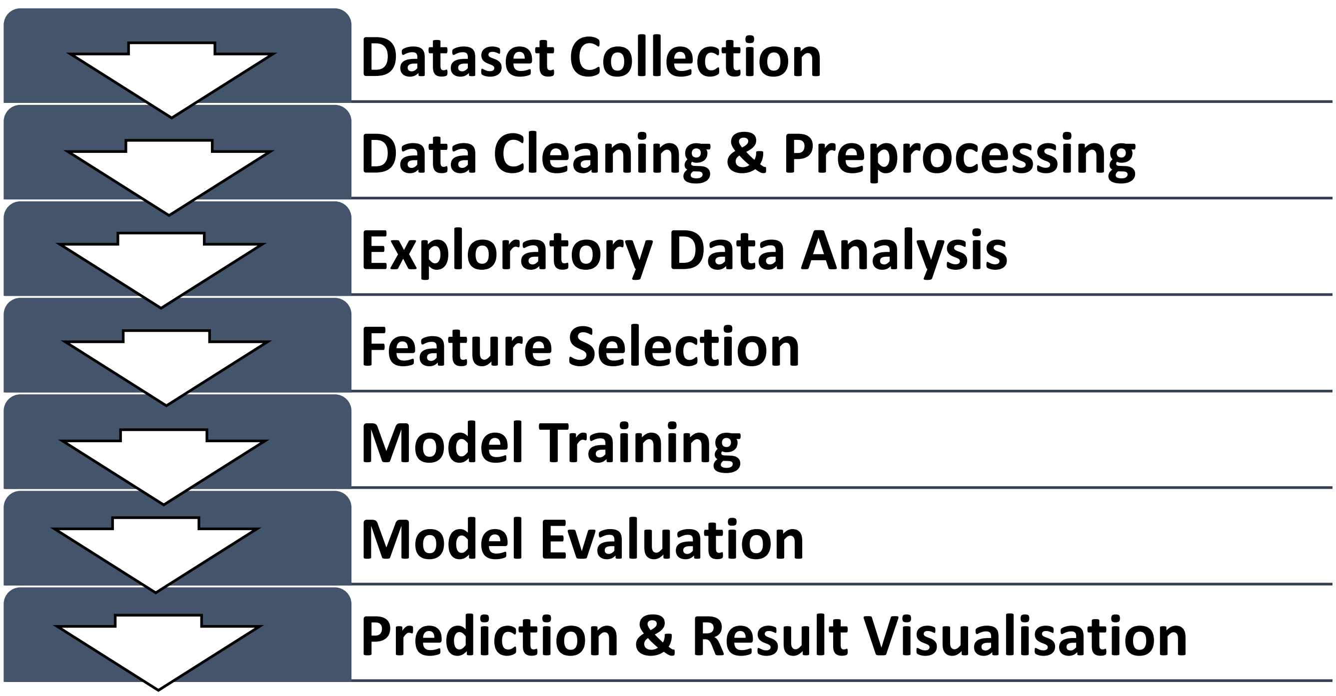
2. Software Requirements

Software	Purpose
Python 3.x	Core programming language
Jupyter Notebook	Interactive development environment
VS Code	Source code editor
Git	Version control system
GitHub	Remote repository hosting

3. Hardware Requirements

Component	Minimum Specification
Processor	Intel Core i3 or equivalent
RAM	4 GB (8 GB recommended)
Storage	10 GB free disk space
Operating System	Windows 10 / Ubuntu 20.04 / macOS 11+

4. Workflow Diagram



5. Database Schema

The dataset fields used as model inputs and the target variable are defined below:

Field Name	Data Type	Description
EmployeeID	Integer	Unique employee identifier
Age	Integer	Age of the employee (years)
Gender	Categorical	Employee gender
Department	Categorical	Organisational department
JobRole	Categorical	Employee's specific job role
MonthlyIncome	Float	Gross monthly salary
YearsAtCompany	Integer	Total tenure at the organisation
JobSatisfaction	Integer	Satisfaction score (1–4 scale)
Attrition	Binary	Target variable: Yes (leaves) / No (stays)

Chapter- 3

Technology Readiness: Implementation & Debugging

Dataset Collection

The Employee Attrition Prediction dataset was obtained from Kaggle. The dataset contains employee-related information such as age, department, monthly income, job role, years at company, job satisfaction, and attrition status. This dataset was used to develop and evaluate machine learning models for predicting employee attrition.

Data Preprocessing

Before building the machine learning models, the dataset was preprocessed to improve data quality and ensure accurate predictions. The dataset was loaded into Python using the Pandas library. Missing values and duplicate records were checked and handled appropriately. Categorical variables were converted into numerical values using encoding techniques. The features and target variables were then separated for model training.

The following preprocessing tasks were performed:

- Handling missing values
- Removing duplicate records
- Encoding categorical variables
- Feature and target variable separation
- Data validation and cleaning

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to understand the structure of the dataset and identify patterns related to employee attrition. Various charts and graphs were created to visualize the distribution of employee data and determine the factors that may influence attrition.

The following visualizations were generated:

- Employee Attrition Distribution
- Department-wise Attrition Analysis
- Monthly Income Distribution
- Job Satisfaction Analysis
- Employee Age Distribution
- Correlation Heatmap

The analysis provided valuable insights into employee behavior and helped in selecting important features for model development.

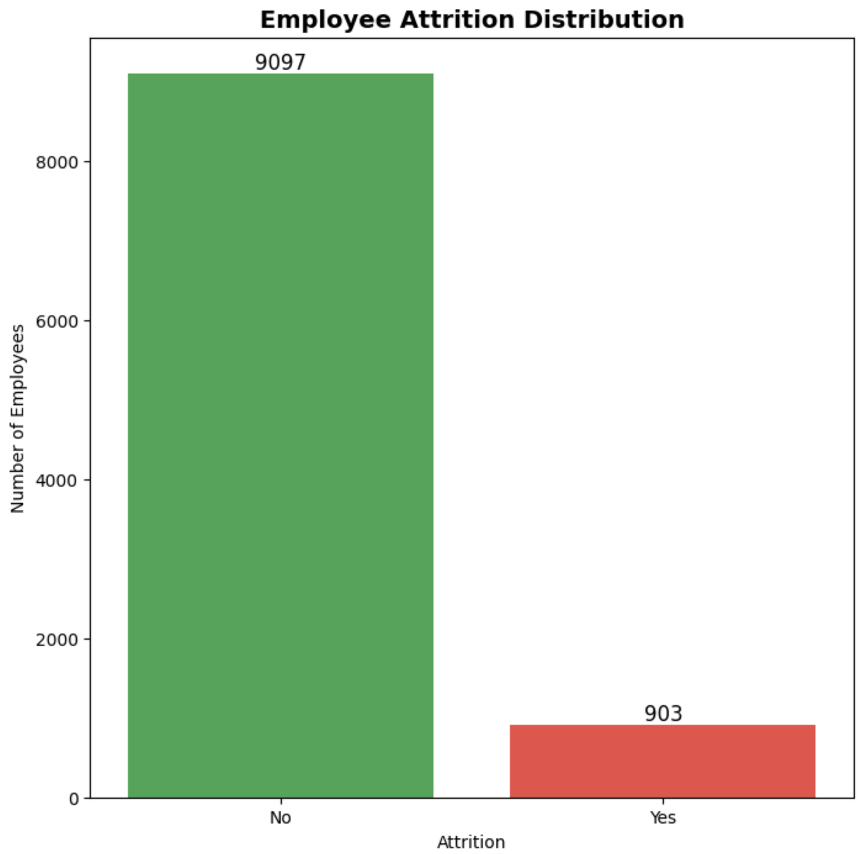


Figure 1_Employee_Attrition_Distribution

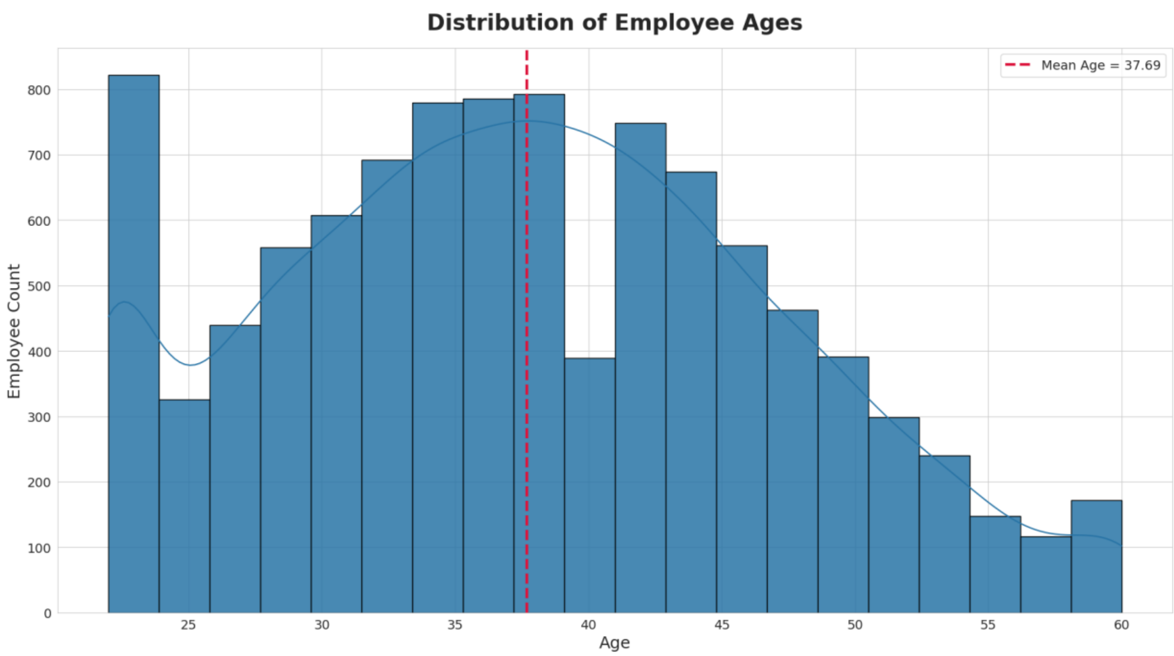


Figure 2_Distribution_of_Employee_Ages

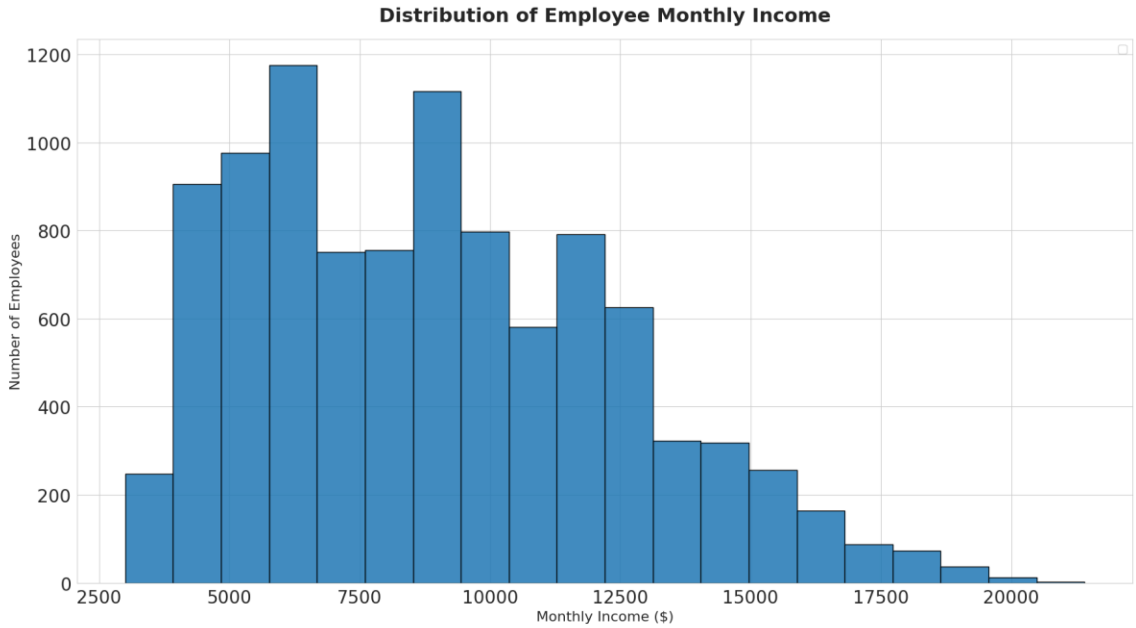


Figure 3_Distribution_of_Employee_Monthly_Income

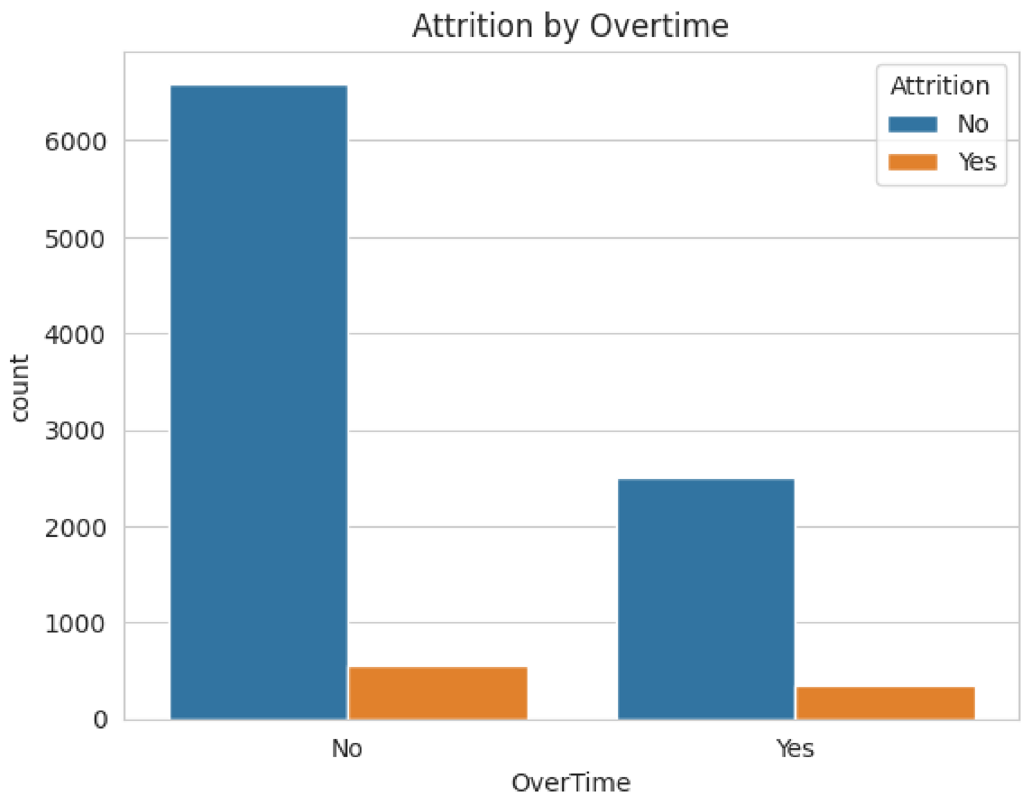


Figure 4_Attrition_by_Overtime

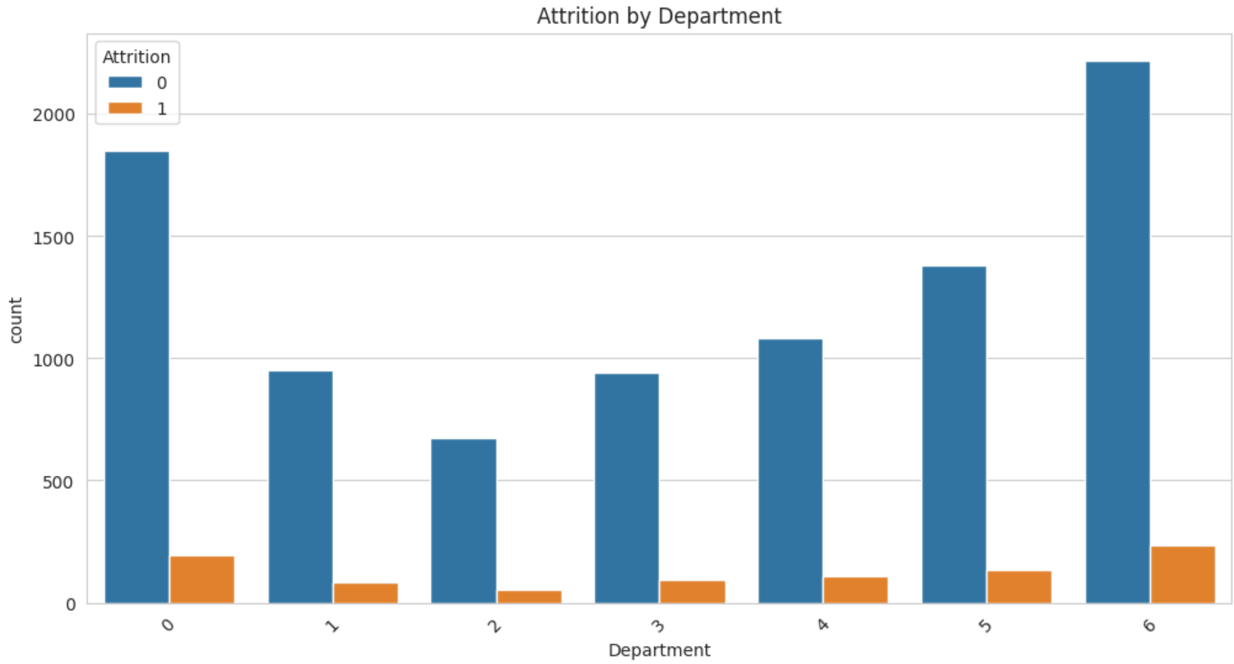


Figure 5_Attrition_by_Department

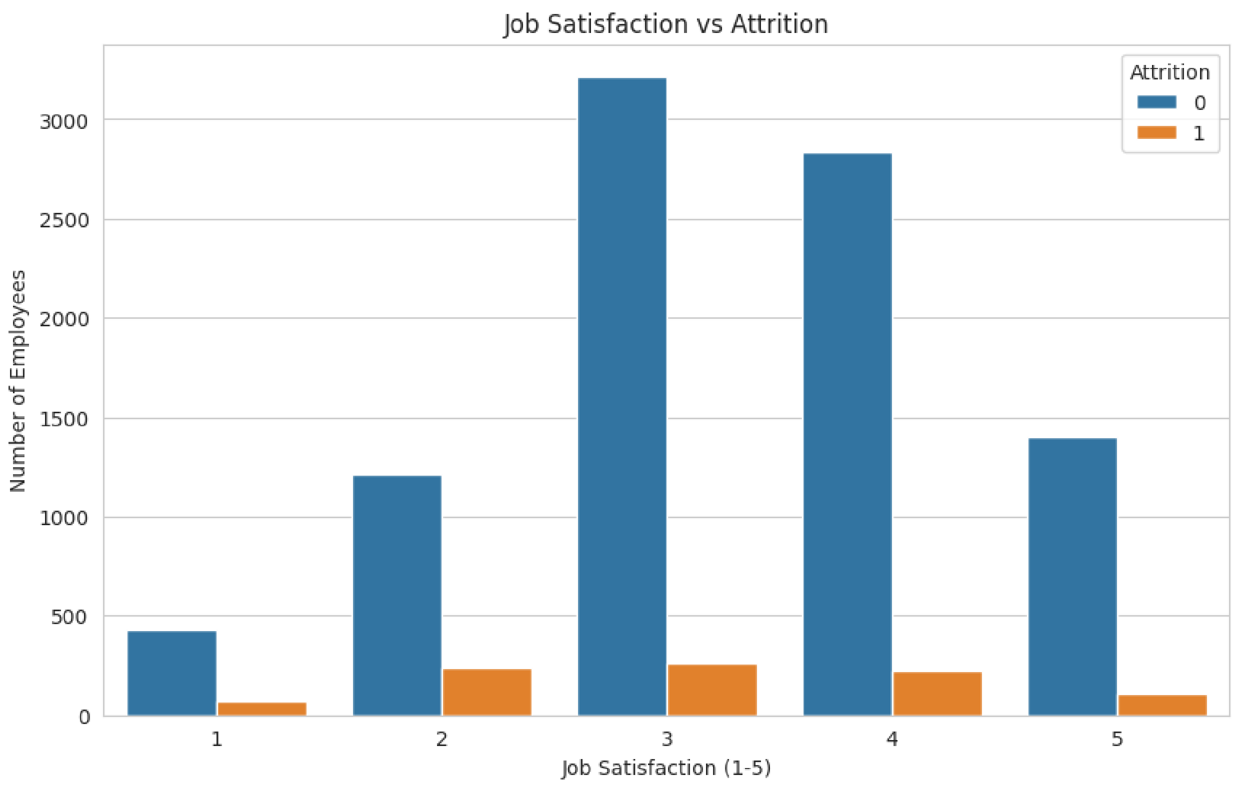


Figure 6_Attrition_by_Department

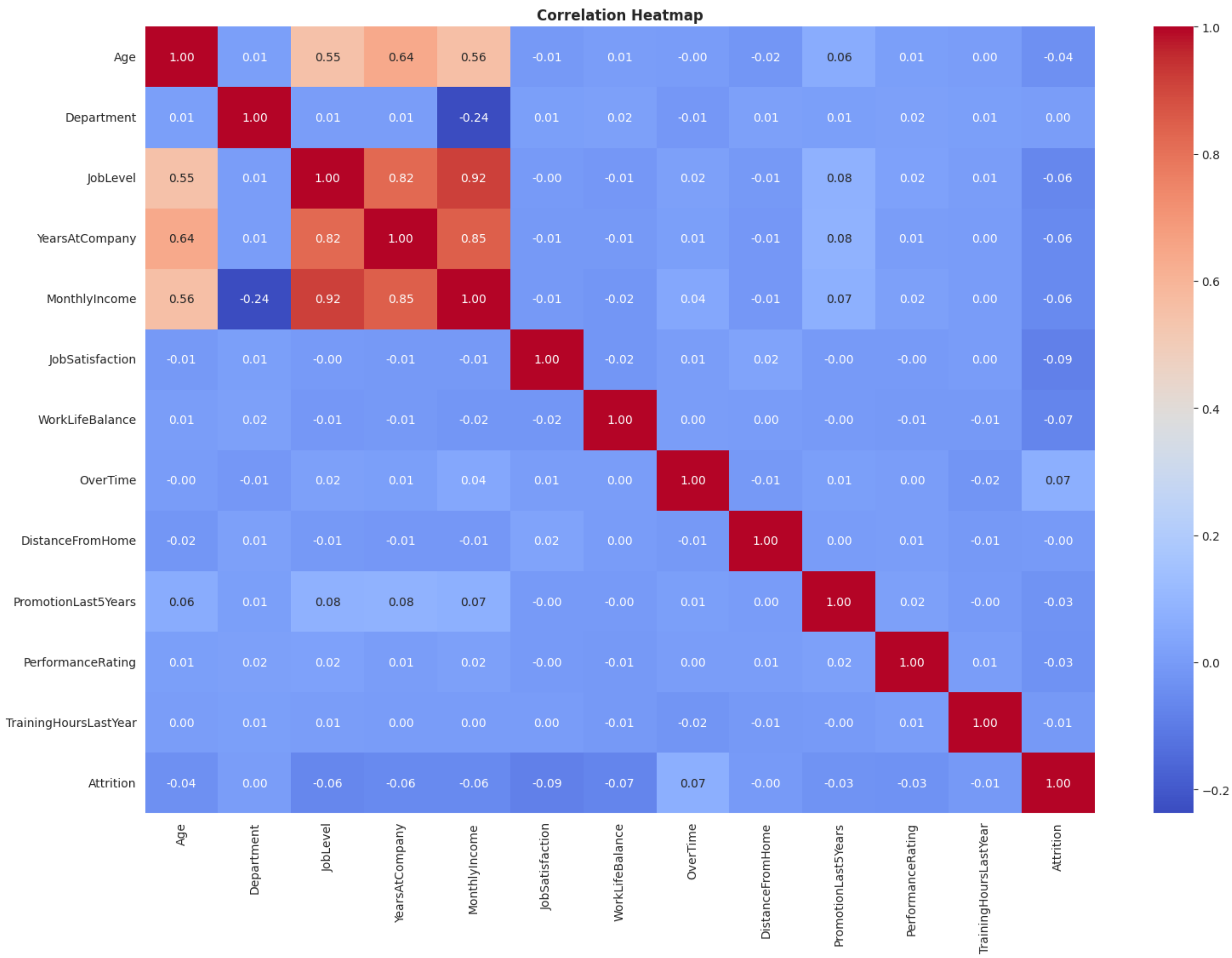


Figure 7_Correlation_Heatmap

Model Development

Two machine learning algorithms were implemented for employee attrition prediction.

Logistic Regression

Logistic Regression was used as a baseline classification algorithm. It predicts whether an employee is likely to stay or leave the organization based on the available employee information.

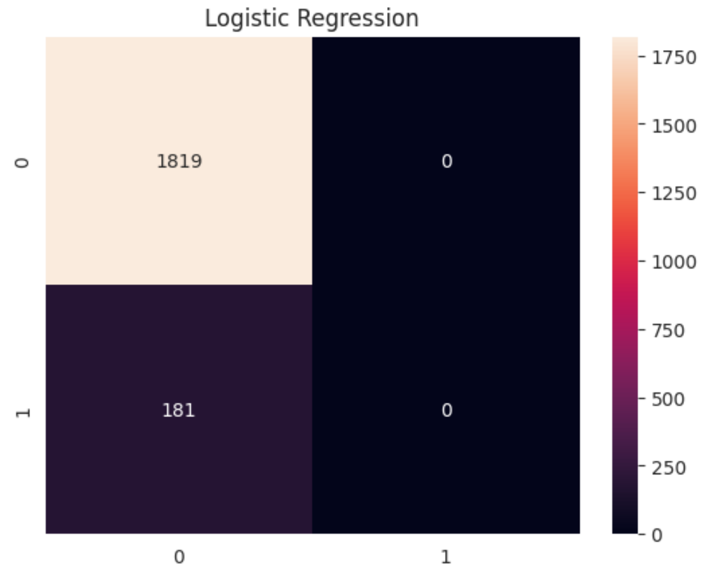


Figure 8_Logistic_Regression

Random Forest Classifier

Random Forest Classifier was implemented as an advanced machine learning model. It combines multiple decision trees to improve prediction accuracy and reduce overfitting.

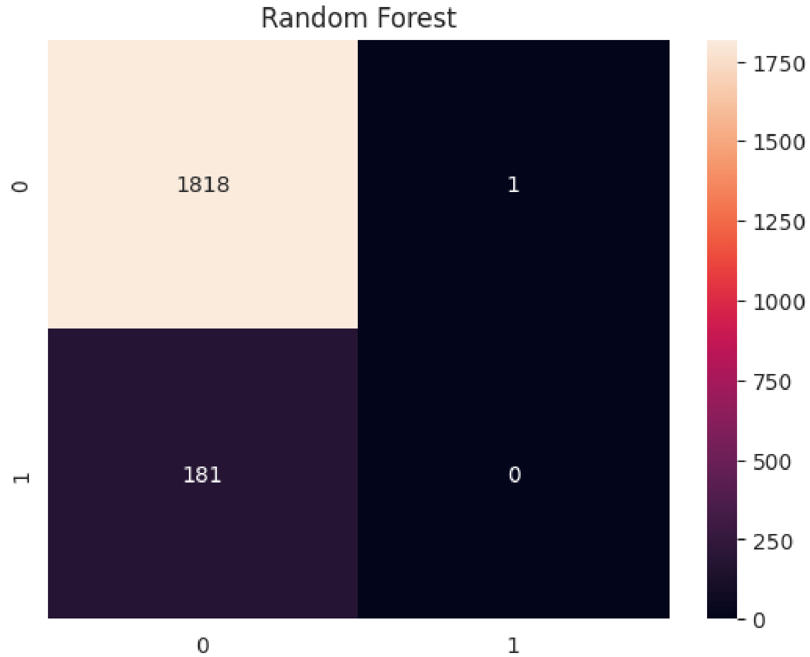


Figure 9_Random_Forest

Model Training

The dataset was divided into training and testing datasets using an 80:20 ratio. The training dataset was used to train the machine learning models, while the testing dataset was used to evaluate their performance on unseen data.

The training process involved:

- Loading the processed dataset
- Splitting the data into training and testing sets
- Training Logistic Regression
- Training Random Forest Classifier
- Generating predictions for evaluation

Model Evaluation

The performance of both machine learning models was evaluated using standard classification metrics. These metrics helped determine the effectiveness of each model in predicting employee attrition.

The following evaluation metrics were used:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

The results obtained from Logistic Regression and Random Forest were compared to identifying the best-performing model.

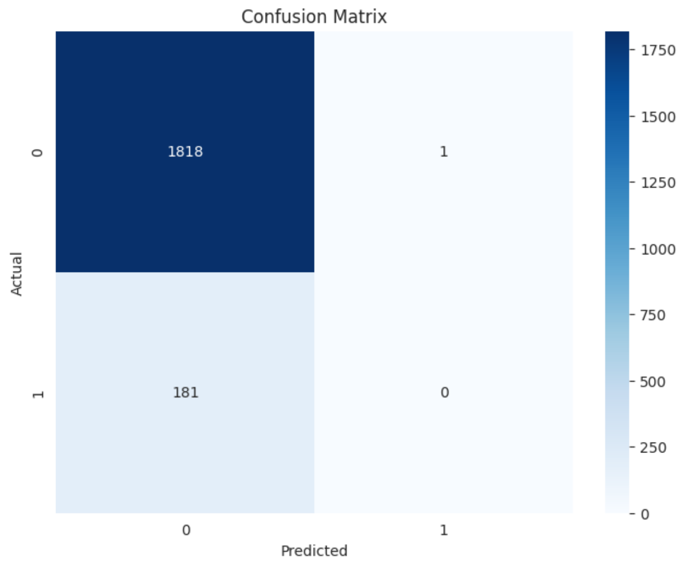


Figure 10_Confusion_matrix

Debugging and Testing

Several testing and debugging activities were performed throughout the implementation process. The project was tested to ensure that all preprocessing, training, and prediction steps were executed correctly.

The following issues were identified and resolved:

- Missing value handling
- Data type conversion errors
- Feature encoding issues
- Prediction validation errors
- Model performance verification

After successful testing and debugging, both machine learning models were able to generate accurate predictions and complete the employee attrition prediction process successfully.

Chapter- 4

Final product: Output screens

The final application presents the following key views:

Screen 1 — Dataset Overview

- Total number of employee records
- Number of input features
- Attrition distribution (Yes vs. No ratio)

```
[6] df = pd.read_csv("Employee_Attrition_DataSet.csv")

[7] print("Dataset Shape:", df.shape)

    df.head()
```

Dataset Shape: (10000, 14)

	EmployeeID	Age	Department	JobLevel	YearsAtCompany	MonthlyIncome	JobSatisfaction	WorkLifeBalance	OverTime	DistanceFromHome
0	EMP0000	42	Sales	3	9	9866	4	1	No	24.5
1	EMP0001	36	Sales	3	7	8958	3	3	No	10.0
2	EMP0002	44	Sales	4	16	11716	4	3	No	4.0
3	EMP0003	53	Finance	4	30	17061	5	3	No	7.1
4	EMP0004	35	HR	2	4	5738	2	3	No	8.4

Next steps: Generate code with df | New interactive sheet

Figure 11_Dataset Overview1

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 13 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   Age                    10000 non-null  int64
 1   Department             10000 non-null  object
 2   JobLevel               10000 non-null  int64
 3   YearsAtCompany         10000 non-null  int64
 4   MonthlyIncome          10000 non-null  int64
 5   JobSatisfaction        10000 non-null  int64
 6   WorkLifeBalance        10000 non-null  int64
 7   OverTime               10000 non-null  object
 8   DistanceFromHome       10000 non-null  float64
 9   PromotionLast5Years    10000 non-null  object
 10  PerformanceRating      10000 non-null  int64
 11  TrainingHoursLastYear  10000 non-null  int64
 12  Attrition              10000 non-null  object
dtypes: float64(1), int64(8), object(4)
memory usage: 1015.8+ KB
```

Figure 12_Dataset Overview 2

Screen 2 — Data Visualisation

- Attrition count plot
- Department-wise attrition breakdown
- Monthly income distribution

```python
plt.figure(figsize=(10,10))

ax = sns.countplot(
    x="Attrition",
    data=df,
    palette=["#4CAF50", "#F44336"]
)

#  numbers on top of bars
for p in ax.patches:
    ax.annotate(
        f'{int(p.get_height())}',
        (p.get_x() + p.get_width() / 2, p.get_height()),
        ha='center',
        va='bottom',
        fontsize=12,

    )

plt.title("Employee Attrition Distribution", fontsize=14, fontweight='bold')
plt.xlabel("Attrition")
plt.ylabel("Number of Employees")

plt.show()
```

Figure 13_Exploratory Data Analysis 1

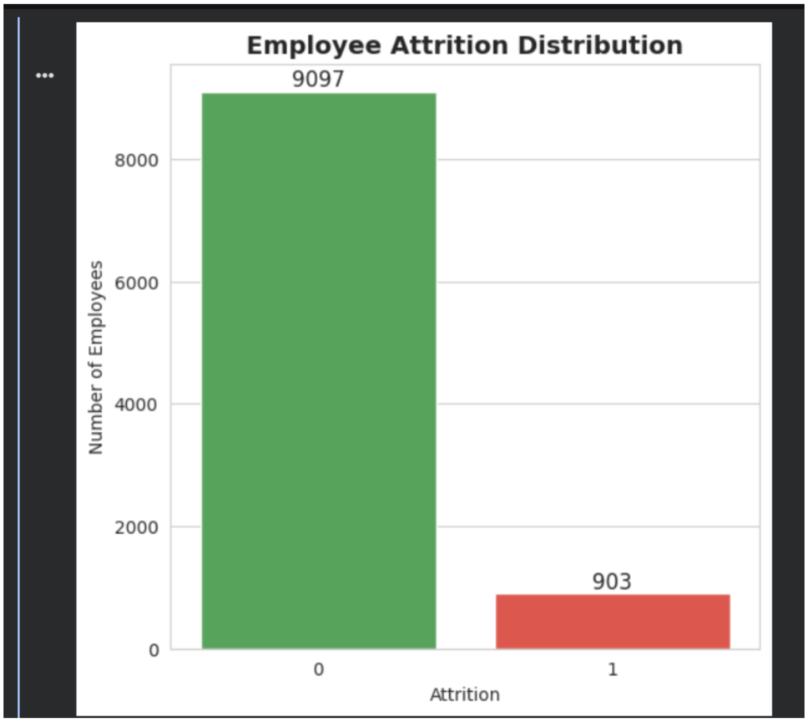


Figure 14_Exploratory Data Analysis_2

```
plt.figure(figsize=(7,7))
sns.countplot(
    x="Department",
    hue="Attrition",
    data=df
)
plt.title("Attrition by Department")
plt.xticks(rotation=45)
plt.show()
```

Figure 15_Exploratory Data Analysis_3

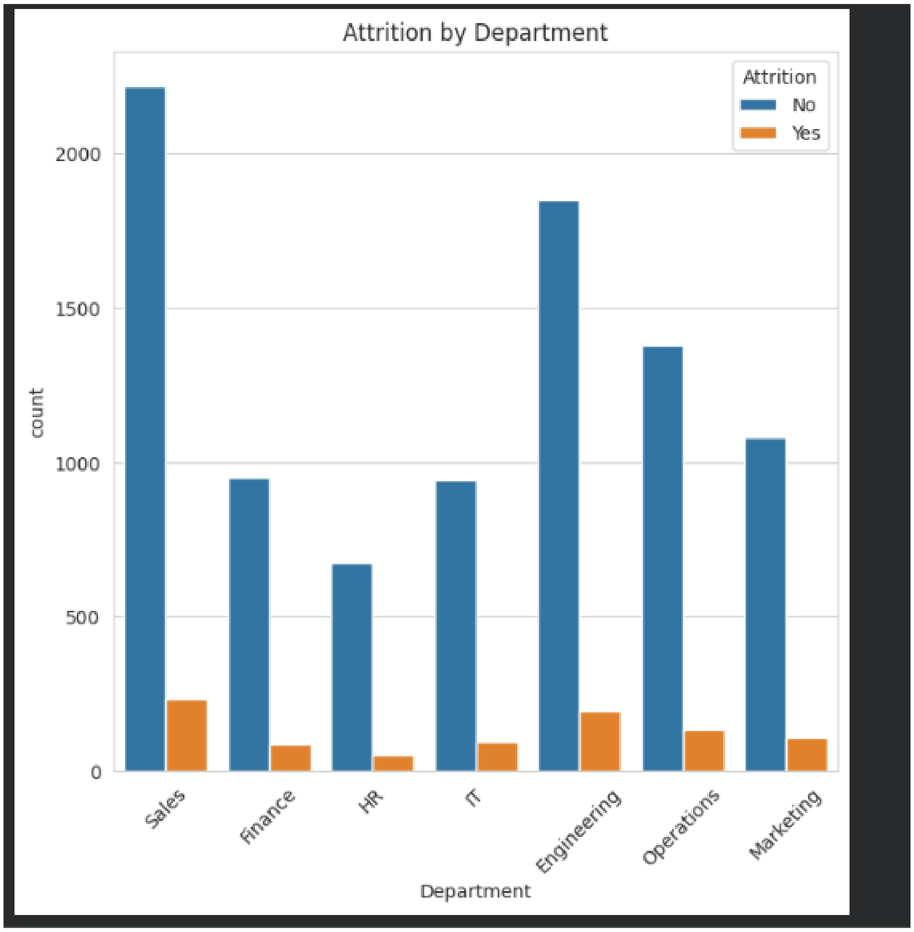


Figure 16_Exploratory Data Analysis _4

```python
import matplotlib.pyplot as plt
import seaborn as sns

# Professional style
sns.set_style("whitegrid")

plt.figure(figsize=(18, 10))

ax = sns.histplot(
    data=df,
    x="MonthlyIncome",
    bins=20,
    color="#1F77B4",
    edgecolor="black",
    alpha=0.85
)
```

Figure 17_Exploratory Data Analysis_5

```python
# Mean income line
mean_income = df["MonthlyIncome"].mean()


# Title and labels
plt.title(
    "Distribution of Employee Monthly Income",
    fontsize=22,
    fontweight="bold",
    pad=20
)

plt.xlabel(
    "Monthly Income ($)",
    fontsize=16
)

plt.ylabel(
    "Number of Employees",
    fontsize=16
)

# Tick formatting
plt.xticks(fontsize=20)
plt.yticks(fontsize=20)
plt.legend(fontsize=13)
plt.tight_layout()
plt.show()
```

Figure 18_Exploratory Data Analysis_6

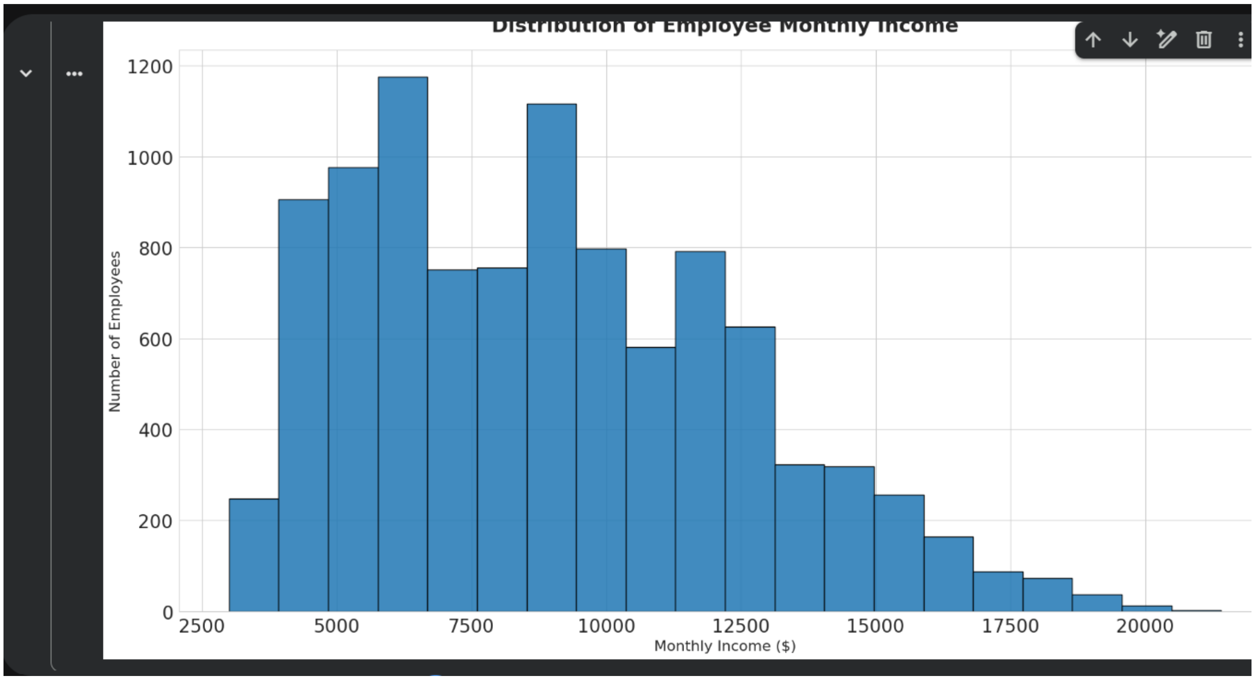


Figure 19_Exploratory Data Analysi_s7

Screen 3 — Model Performance Metrics

Metric	Description
Accuracy Score	Overall percentage of correct predictions
Precision Score	Proportion of predicted positives that are truly positive
Recall Score	Proportion of actual positives correctly identified
Confusion Matrix	Visual breakdown of true/false positives and negatives

```python
X = df.drop("Attrition", axis=1)

y = df["Attrition"]
```

```python
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
```

```python
print(X_train.shape)
print(X_test.shape)
```

```
(8000, 12)
(2000, 12)
```

```python
lr = LogisticRegression(max_iter=2000)

lr.fit(X_train, y_train)

lr_pred = lr.predict(X_test)
```

Figure 20_Model Training_1

```python
print("Accuracy:",
      accuracy_score(y_test, lr_pred))
```

```
Accuracy: 0.9095
```

```python
print(classification_report(
    y_test,
    lr_pred
))
```

```
              precision    recall  f1-score   support

           0       0.91      1.00      0.95      1819
           1       0.00      0.00      0.00       181

    accuracy                           0.91      2000
   macro avg       0.45      0.50      0.48      2000
weighted avg       0.83      0.91      0.87      2000
```

Figure 21_Model Training_2

```
[91]
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=58,
    stratify=y
)
```

```
[92]
print(X_train.shape)
print(X_test.shape)
```

```
(8000, 12)
(2000, 12)
```

```
[93]
lr = LogisticRegression(max_iter=2000)

lr.fit(X_train, y_train)

lr_pred = lr.predict(X_test)
```

```
[94]
print("Accuracy:",
      accuracy_score(y_test, lr_pred))
```

```
Accuracy: 0.9095
```

Figure 22_Model Training _3

```
print(classification_report(
    y_test,
    lr_pred
))
```

```
              precision    recall  f1-score   support

           0       0.91      1.00      0.95      1819
           1       0.00      0.00      0.00       181

    accuracy                           0.91      2000
   macro avg       0.45      0.50      0.48      2000
weighted avg       0.83      0.91      0.87      2000
```

```
cm = confusion_matrix(y_test, lr_pred)

sns.heatmap(
    cm,
    annot=True,
    fmt="d"
)

plt.title("Logistic Regression")
plt.show()
```

Figure 23_Model Training_4

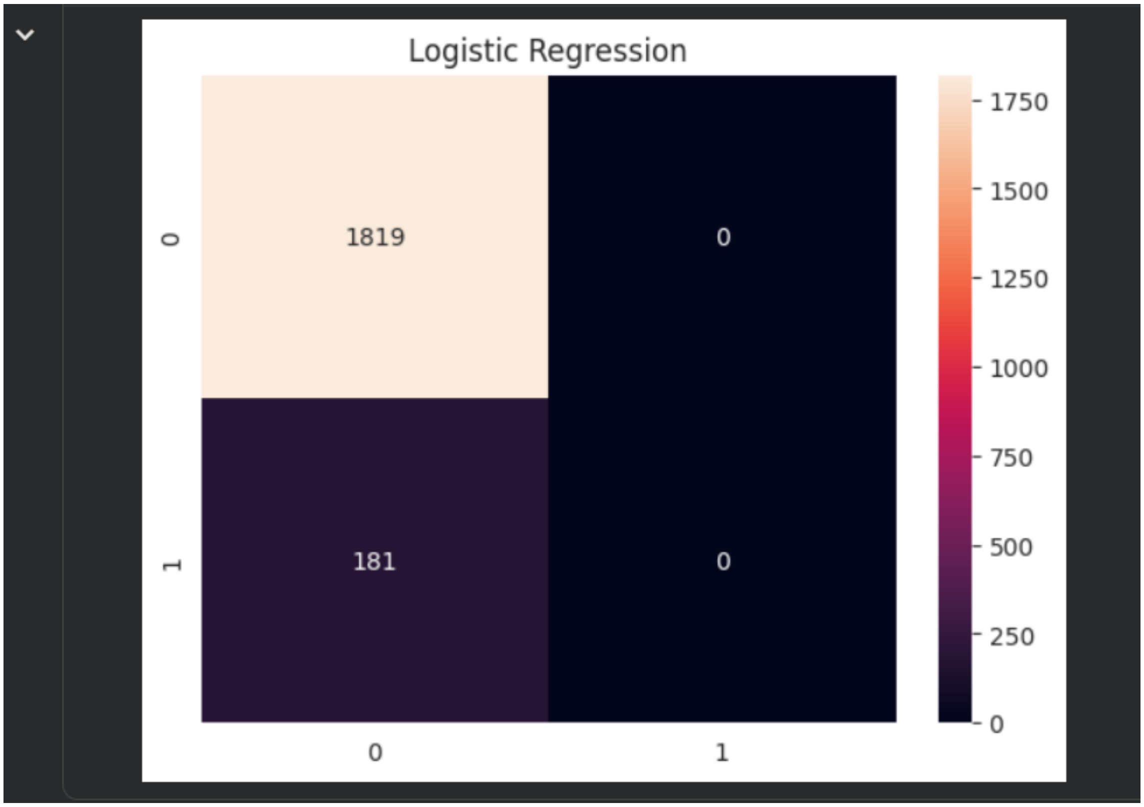


Figure 24_Model Training_5

```
rf = RandomForestClassifier(
    n_estimators=200,
    random_state=42
)

rf.fit(X_train, y_train)

rf_pred = rf.predict(X_test)
```

```
print("Accuracy:",
      accuracy_score(y_test, rf_pred))
```

```
Accuracy: 0.909
```

```
print(classification_report(
    y_test,
    rf_pred
))
```

Figure 25_Model Training_6

```
              precision    recall  f1-score   support

           0       0.91      1.00      0.95      1819
           1       0.00      0.00      0.00       181

    accuracy                           0.91      2000
   macro avg       0.45      0.50      0.48      2000
weighted avg       0.83      0.91      0.87      2000
```

```
[100]
cm = confusion_matrix(y_test, rf_pred)

sns.heatmap(
    cm,
    annot=True,
    fmt="d"
)

plt.title("Random Forest")
plt.show()
```

Figure 26_Model Training_7

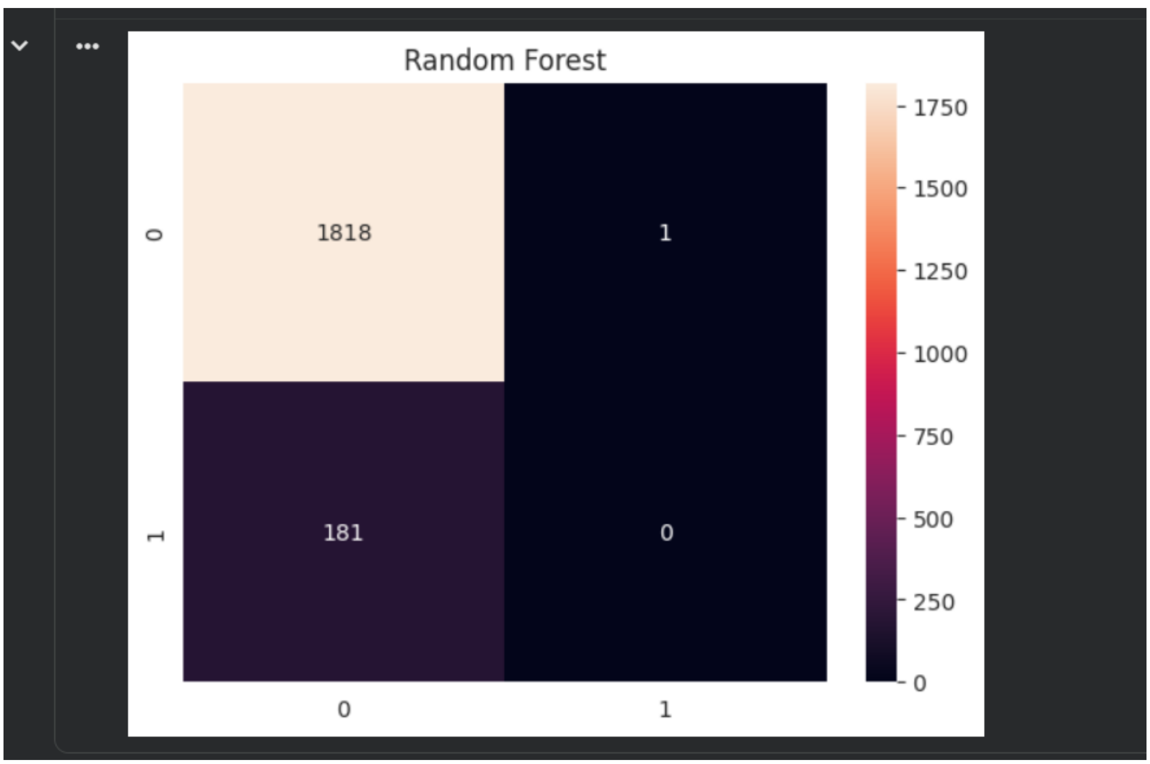


Figure 27_Model Training_8

Screen 4 — Prediction Interface

Users input employee details via a form; the model returns:

Employee Likely to Stay
Employee Likely to Leave

```python
sample_employee = X.iloc[[0]]

prediction = loaded_model.predict(
    sample_employee
)

print(prediction)
```

```
[0]
```

```python
if prediction[0] == 1:
    print("Employee Likely to Leave")
else:
    print("Employee Likely to Stay")
```

```
Employee Likely to Stay
```

```python
from sklearn.metrics import accuracy_score

predictions = rf.predict(X_test)

accuracy = accuracy_score(y_test, predictions)

print("Accuracy:", accuracy)
```

```
Accuracy: 0.909
```

Figure 28_Model_Evaluation_Results_1

```
sample_employee = X_test.iloc[[0]]

prediction = rf.predict(sample_employee)

print("Prediction:", prediction)
print("Actual:", y_test.iloc[0])
```

```
Prediction: [0]
Actual: 0
```

```
if prediction[0] == 1:
    print("Employee likely to leave")
else:
    print("Employee likely to stay")
```

```
Employee likely to stay
```

Figure 29_Model_Evaluation_Results_2

```
from sklearn.metrics import confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

cm = confusion_matrix(y_test, predictions)

plt.figure(figsize=(8,6))

sns.heatmap(
    cm,
    annot=True,
    fmt='d',
    cmap='Blues'
)

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")

plt.show()
```

Figure 30_Model_Evaluation_Results_3

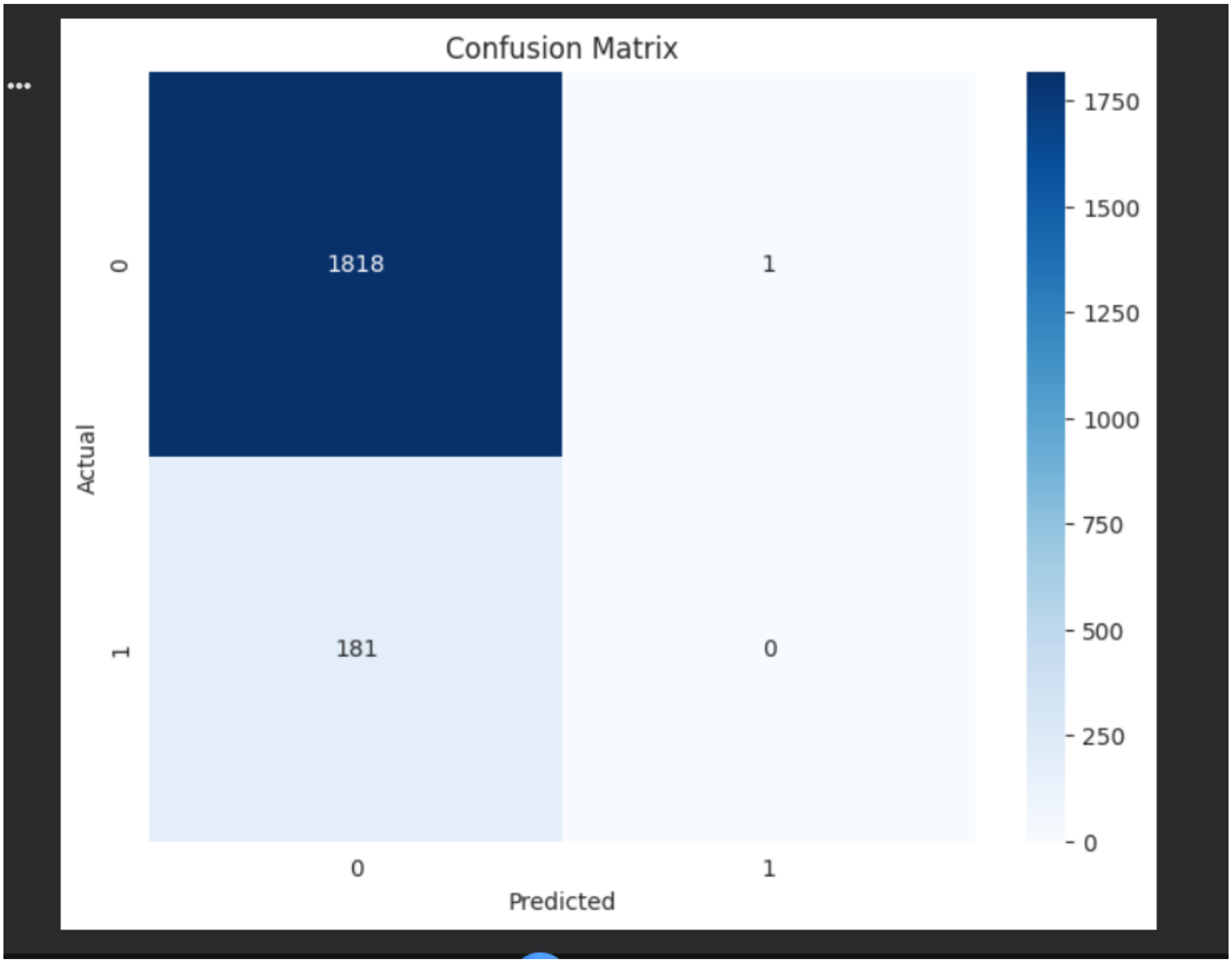


Figure 31_Model_Evaluation_Results_4

Chapter- 5

GitHub & Deployment details

GitHub Repository

The Employee Attrition Prediction project was developed using Python and Jupyter Notebook. The source code, dataset, documentation, and project files can be stored and managed using GitHub for version control and collaboration.

Repository Structure

```
Employee-Attrition-Prediction/
├── dataset/
├── employee_attrition.csv
├── Employee_Attrition_Prediction.ipynb
├── pics
├── README.md
└── requirements.txt
```

Deployment Details

The project was implemented and tested in a local development environment using Jupyter Notebook and Python.

Currently, the project has not been deployed to a cloud platform. All data preprocessing, model training, evaluation, and prediction tasks were executed successfully on a local machine.

https://github.com/abdalleofficial/Predicting-Employee-Attrition---Machine-Learning-Project-/tree/main

Chapter- 6

Conclusion & Discussions

Employee attrition prediction represents a high-value application of machine learning within the domain of Human Resource Management. This project demonstrates how structured employee data can be transformed into actionable predictive intelligence, enabling organisations to move from reactive to proactive talent management.

The developed model successfully analyses employee attributes and classifies individuals according to their likelihood of departure. Beyond mere prediction, the model surfaces the underlying drivers of attrition, equipping HR professionals with the contextual information needed to design targeted retention initiatives.

With further refinement, real-time integration, and deployment as an end-user application, this system has the potential to become an indispensable component of modern workforce analytics platforms.

Chapter- 7

Bibliography & References

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[3] Pandas Documentation — https://pandas.pydata.org/

[4] NumPy Documentation — https://numpy.org/

[5] Seaborn Documentation — https://seaborn.pydata.org/

[6] Python Official Documentation — https://docs.python.org/3/

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Appendix

Appendix A: Dataset Information

Dataset Name: Employee Attrition Prediction Dataset

Source: Kaggle

Dataset Link: https://www.kaggle.com/datasets/ziya07/employee-attrition-prediction-dataset

Description: The dataset contains employee-related information used to predict employee attrition. It includes demographic, professional, and satisfaction-related features.

Appendix B: Software and Tools Used

- Python 3.x
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn
- GitHub

Appendix C: Machine Learning Models

Logistic Regression

Logistic Regression was used as a baseline classification model to predict employee attrition.

Random Forest Classifier

Random Forest was used to improve prediction performance and compare results with Logistic Regression.

Appendix D: Sample Code

Loading Dataset

```
import pandas as pd

df = pd.read_csv("employee_attrition.csv")
```

Logistic Regression

```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression()
model.fit(X_train, y_train)
```

Random Forest

```
from sklearn.ensemble import RandomForestClassifier

rf = RandomForestClassifier()
rf.fit(X_train, y_train)
```

Appendix E: Project Screenshots

- Dataset Overview
- Exploratory Data Analysis Graphs
- Model Training Results
- Confusion Matrix
- Prediction Results

(Insert screenshots here)

Appendix F: Abbreviations

Abbreviation	Meaning
EDA	Exploratory Data Analysis
ML	Machine Learning
AI	Artificial Intelligence
HR	Human Resources
RF	Random Forest
LR	Logistic Regression

Appendix G: Future Enhancements

- Development of a web application
- Real-time employee prediction system
- Advanced machine learning models
- HR analytics dashboard
- Cloud deployment